LMECA2660 project: Simulation of inviscid transonic flows past a lens-shape airfoil, steady and unsteady

G. Winckelmans, G. Querinjean, G. Poncelet

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We consider the Euler equations for compressible flow in 2-D:

$$\frac{\partial \mathbf{s}}{\partial t} + \frac{\partial}{\partial x}\mathbf{F}_x(\mathbf{s}) + \frac{\partial}{\partial y}\mathbf{F}_y(\mathbf{s}) = 0 \tag{1}$$

with

$$\mathbf{s} = \begin{pmatrix} s_1 \\ s_2 \\ s_3 \\ s_4 \end{pmatrix} = \begin{pmatrix} \rho \\ \rho\, u \\ \rho\, v \\ \rho\, U_0 \end{pmatrix} \tag{2}$$

$$\mathbf{F}_x = \begin{pmatrix} F_{x,1} \\ F_{x,2} \\ F_{x,3} \\ F_{x,4} \end{pmatrix} = \begin{pmatrix} \rho\, u \\ \rho\, u\, u + p \\ \rho\, v\, u \\ \rho\, H_0\, u \end{pmatrix} \qquad \mathbf{F}_y = \begin{pmatrix} F_{y,1} \\ F_{y,2} \\ F_{y,3} \\ F_{y,4} \end{pmatrix} = \begin{pmatrix} \rho\, v \\ \rho\, u\, v \\ \rho\, v\, v + p \\ \rho\, H_0\, v \end{pmatrix} \tag{3}$$

with $U_0 = U + \frac{(u^2+v^2)}{2}$ the total internal energy where U is the internal energy ($dU = c_v\, dT$ for a perfect gas), and $H_0 = H + \frac{(u^2+v^2)}{2} = U_0 + \frac{p}{\rho}$ the total enthalpy where $H = U + \frac{p}{\rho}$ is the enthalpy (with $dH = c_p\, dT$).

The fluid is dry air, with $R = c_p - c_v = 287.1$ J/kg K and $\gamma = \frac{c_p}{c_v} = 1.40$. The geometry corresponds to a wide channel, with a thin airfoil placed in the middle. The shape of the airfoil is a lens, formed using two arcs of circle. The airfoil chord is $c = 1.0$ m and the radius of the circle is $R = 3\,c$. We then obtain than the airfoil thickness is $e \simeq 0.08392\,c$. The origin of the $x-y$ coordinate system is located at the leading edge of the airfoil. The total height of the channel is $H = 4\,c$. The total length of the computational domain is L, and $\frac{L}{c}$ is left to your choice. Its part upstream of the airfoil is noted L_u, and $\frac{L_u}{c}$ is also left to your choice. Its part downstream of the airfoil is noted L_d, thus: $L_d = L-(L_u+c)$.

When the airfoil angle of attack is zero and the grid is fixed (i.e., no airfoil movement), the problem is symetric relative to the center of the channel. Hence, it can be solved for the upper part only, $0 \leq y \leq H/2$: this corresponds to the case shown in Fig. 2.

We will integrate in time the Euler equations written in conservative form in the computational domain. We will use second order centered finite differences for the spatial discretisation, and the second order two-step Mac-Cormack scheme for the temporal

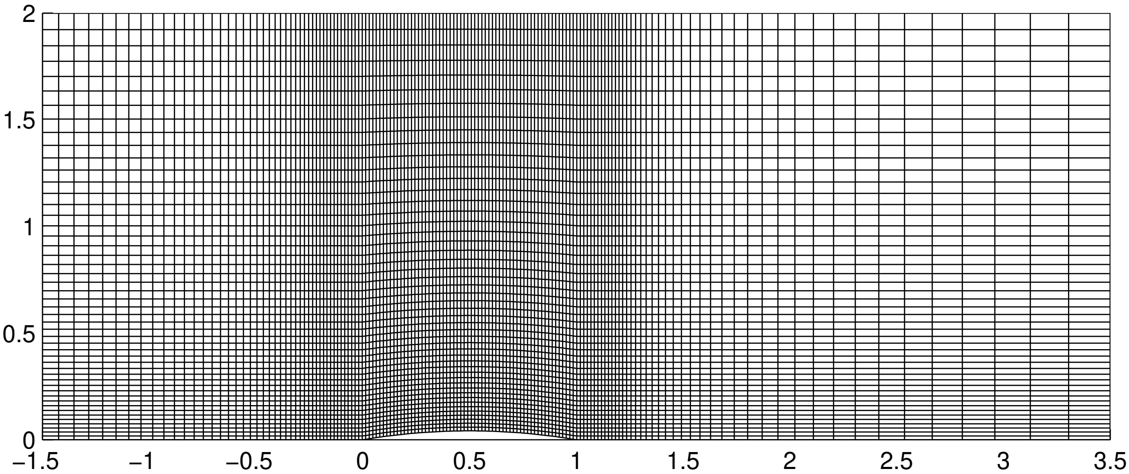

Figure 1: illustration of the domain for steady flow past a lens-shape airfoil without angle of attack. A stretched grid with $\frac{L}{c} = 5$ and $\frac{L_u}{c} = \frac{3}{2}$ is also shown.

integration.

When simulating flows with a moving airfoil, the grid also deforms in time: this must be taken into account. If the time step is taken constant in time, then $t^n = n\,\Delta t$. Otherwise, it is a function of n: $t^n = t(n)$.,For the general case with a moving airfoil, the Jacobian matrix J of the transformation $(t, i, , j) \to (t, x, y)$ is

$$J = \begin{pmatrix} 1 & 0 & 0 \\ \frac{\partial x}{\partial t} & \frac{\partial x}{\partial i} & \frac{\partial x}{\partial j} \\ \frac{\partial y}{\partial t} & \frac{\partial y}{\partial i} & \frac{\partial y}{\partial j} \end{pmatrix}, \tag{4}$$

and we use the notation $\mathcal{D} = \left(\frac{\partial x}{\partial i}\frac{\partial y}{\partial j} - \frac{\partial x}{\partial j}\frac{\partial y}{\partial i}\right) > 0$ for its determinant (= the Jacobian of the transformation).

The Jacobian matrix J^{-1} of the inverse transformation $(t, x, y) \to (t, i, , j)$ is

$$J^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ \frac{\partial i}{\partial t} & \frac{\partial i}{\partial x} & \frac{\partial i}{\partial y} \\ \frac{\partial j}{\partial t} & \frac{\partial j}{\partial x} & \frac{\partial j}{\partial y} \end{pmatrix}, \tag{5}$$

and we use the notation $\mathcal{D}^{-1} = \left(\frac{\partial i}{\partial x}\frac{\partial j}{\partial y} - \frac{\partial i}{\partial y}\frac{\partial j}{\partial x}\right) > 0$ for its determinant. This notation is justified by the fact that it is the inverse of $\mathcal{D}$: $\mathcal{D}^{-1} = \frac{1}{\left(\frac{\partial x}{\partial i}\frac{\partial y}{\partial j} - \frac{\partial x}{\partial j}\frac{\partial y}{\partial i}\right)} = \frac{1}{\mathcal{D}}$.

When it is J that is known for each (t, i, j), then J^{-1} is obtained (using matrix inversion) as

$$J^{-1} = \frac{1}{\mathcal{D}} \begin{pmatrix} \mathcal{D} & 0 & 0 \\ \frac{\partial y}{\partial t}\frac{\partial x}{\partial j} - \frac{\partial x}{\partial t}\frac{\partial y}{\partial j} & \frac{\partial y}{\partial j} & -\frac{\partial x}{\partial j} \\ \frac{\partial x}{\partial t}\frac{\partial y}{\partial i} - \frac{\partial y}{\partial t}\frac{\partial x}{\partial i} & -\frac{\partial y}{\partial i} & \frac{\partial x}{\partial i} \end{pmatrix}. \tag{6}$$

When it is J^{-1} that is known for each (t,i,j), then J is obtained as

$$J = \frac{1}{\mathcal{D}^{-1}} \begin{pmatrix} \frac{1}{\mathcal{D}^{-1}} & 0 & 0 \\ \frac{\partial j}{\partial t}\frac{\partial i}{\partial y} - \frac{\partial i}{\partial t}\frac{\partial j}{\partial y} & \frac{\partial j}{\partial y} & -\frac{\partial i}{\partial y} \\ \frac{\partial i}{\partial t}\frac{\partial j}{\partial x} - \frac{\partial j}{\partial t}\frac{\partial i}{\partial x} & -\frac{\partial j}{\partial x} & \frac{\partial i}{\partial x} \end{pmatrix} . \tag{7}$$

The Euler equations in conservative form in the computational domain are then obtained as (see course notes):

$$\frac{\partial(\mathcal{D}\,\mathbf{s})}{\partial t} + \frac{\partial \tilde{\mathbf{F}}_i}{\partial i} + \frac{\partial \tilde{\mathbf{F}}_j}{\partial j} = 0 \tag{8}$$

with

$$\begin{aligned} \tilde{\mathbf{F}}_i &= \mathcal{D}\left(\frac{\partial i}{\partial t}\mathbf{s} + \frac{\partial i}{\partial x}\mathbf{F}_x + \frac{\partial i}{\partial y}\mathbf{F}_y\right) \\ &= \left(\frac{\partial y}{\partial t}\frac{\partial x}{\partial j} - \frac{\partial x}{\partial t}\frac{\partial y}{\partial j}\right)\mathbf{s} + \frac{\partial y}{\partial j}\mathbf{F}_x - \frac{\partial x}{\partial j}\mathbf{F}_y \; , \\ \tilde{\mathbf{F}}_j &= \mathcal{D}\left(\frac{\partial j}{\partial t}\mathbf{s} + \frac{\partial j}{\partial x}\mathbf{F}_x + \frac{\partial j}{\partial y}\mathbf{F}_y\right) \\ &= \left(\frac{\partial x}{\partial t}\frac{\partial y}{\partial i} - \frac{\partial y}{\partial t}\frac{\partial x}{\partial i}\right)\mathbf{s} - \frac{\partial y}{\partial i}\mathbf{F}_x + \frac{\partial x}{\partial i}\mathbf{F}_y \; . \end{aligned} \tag{9}$$

In order to stabilize the numerical time integration, and also capture numerically discontinuities (= shocks), we add an artificial diffusion term to equations. It is also performed in the computational domain:

$$\frac{\partial(\mathcal{D}\,\mathbf{s})}{\partial t} = -\left[\frac{\partial \tilde{\mathbf{F}}_i}{\partial i} + \frac{\partial \tilde{\mathbf{F}}_j}{\partial j}\right] + \left[\frac{\partial}{\partial i}\left(\epsilon\,\frac{\partial \mathbf{s}}{\partial i}\right) + \frac{\partial}{\partial j}\left(\epsilon\,\frac{\partial \mathbf{s}}{\partial j}\right)\right] \tag{10}$$

with ϵ of dimensions $[L^2/T]$ and chosen based on the gradients of the local velocity (so that it is large in regions with fast velocity variation and small in regions with slow velocity variation). For instance, we use

$$\epsilon_{i+1/2,j} = \epsilon_{min} + C\;c\sqrt{(u_{i+1,j} - u_{i,j})^2 + (v_{i+1,j} - v_{i,j})^2} \; , \tag{11}$$

etc., with C a dimensionless coefficient to be calibrated to obtain "satisfactory results". Note that we also added a minimum value ϵ_{min}, also of units $[L^2/T]$ and taken as constant, to enforce a minimum diffusion everywhere. The time integration of the added artificial diffusion term will be performed using the Euler scheme.

1. Express the components of $\mathbf{F}_x$ and $\mathbf{F}_y$ as a function of the components of $\mathbf{s}$.

2. Obtain the detailed equations for the metric, as you will need to evaluate them efficiently.

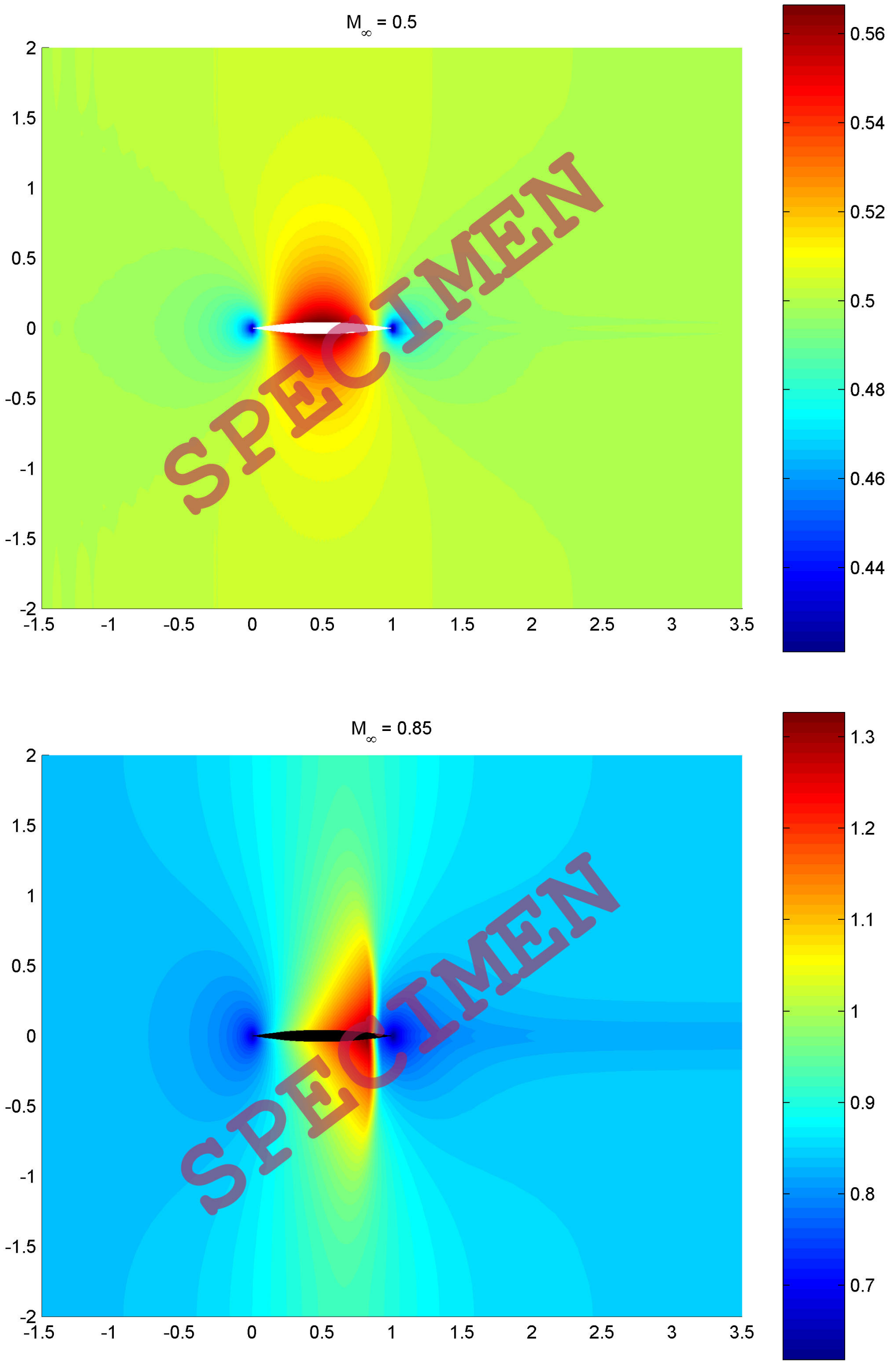

Figure 2: Example of Mach number color plot for steady flow past a lens-shape airfoil without angle of attack at $M_\infty = 0.50$ (top) and at $M_\infty = 0.85$ (bottom).

3. The flow is subsonic at inlet, thus we must impose three conditions and extrapolate one from the nearby grid point. We impose $v = 0$, the total enthalpy $H_0 = H_{0,\infty}$ and the entropy $S = S_\infty$, wich is equivalent to imposing that $p/p_\infty = (\rho/\rho_\infty)^\gamma$.

   The upstream conditions are determined by the Mach number $M_\infty = 0.85$, together with the temperature $T_\infty = 300$ K and pressure $p_\infty = 1$ bar of the atmosphere (assuming we fly close to sea level). The total temperature and the total pressure are thus obtained as:

$$\frac{T_{0,\infty}}{T_\infty} = 1 + \frac{(\gamma-1)}{2} M_\infty^2 \, , \qquad \frac{p_{0,\infty}}{p_\infty} = \left(\frac{T_{0,\infty}}{T_\infty}\right)^{\frac{\gamma}{(\gamma-1)}} \, . \tag{12}$$

4. The flow is subsonic at outlet, thus we must impose one condition and extrapolate three from the nearby grid point. We impose $p = p_\infty$ and we extrapolate s_2, s_3 and s_4.

5. On slip walls, we assume a zero normal derivative for all quantities, as evaluated in the computational domain. We also impose that the velocity vector is parallel to the wall. For a horizontal wall, such as the channel boundaries, this simply translates into $s_3 = 0$ at the boundary. For curved boundaries, such as the airfoil, we simply tilt the local velocity vector after its update to keep it tangent to the wall (i.e., with zero normal velocity component).

At first, you will simulate the steady flow at $M_\infty = 0.85$ past the airfoil put horizontally (no angle of attack) and centered in the middle of the channel:

- Produce the two grids and compute the terms of the metric. Here, you can store them (as they don't vary in time here).

- To illustrate the process of convergence, the histogram of the residue for each variable s_k is evaluated using

$$R_k(n) = \int \left(s_k^n - s_k^{n-1}\right)^2 \, dx\, dy \simeq \sum_{i,j} \left(s_{k_{i,j}}^n - s_{k_{i,j}}^{n-1}\right)^2 \mathcal{D}_{i,j} \, . \tag{13}$$

  As they vary by many orders of magnitude, you should plot $\log_{10}\left(\frac{R_k(n)}{R_k(0)}\right)$ as a function of n.

- At convergence, produce color plots of interesting results:
  - the Mach number, M;
  - the pressure coefficient $C_p = \frac{(p-p_\infty)}{\rho_\infty U_\infty^2/2} = \frac{(p/p_\infty-1)}{\gamma M_\infty^2/2}$;
  - the normalized total enthalpy $\frac{H_0}{H_{0,\infty}}$ (conserved for steady flows, even with shock(s); thus a good indicator of the quality of the simulation);
  - the normalized total pressure, $\frac{p_0}{p_{0,\infty}}$ (conserved for steady flows without shock; not conserved for steady flow with shock(s));
  - the normalized entropy $\frac{(S-S_\infty)}{R} = \frac{1}{(\gamma-1)} \log\left(\frac{p/p_\infty}{(\rho/\rho_\infty)^\gamma}\right)$;

- the normalized vorticity $\frac{\omega c}{U_\infty}$ with $\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$.

Also provide the plots of M and C_p on the upper and lower walls, and on $f_{m^+}(x)$ and $f_{m^+}(x)$. Also compute the drag coefficient.

Advice: Before simulating the transonic flow case with $M_\infty = 0.85$, it is recommended to simulate the case with $M_\infty = 0.50$, which corresponds to a flow that remains subsonic everywhere; thus, it is isentropic and symmetric in x relative to the middle of the airfoil.

Next you will simulate the steady flow at $M_\infty = 0.85$ past the airfoil put at some angle of attack and centered in the middle of the channel.

Finally, you will simulate one or more cases of unsteady flows, of your choice. There are various possibilities, still maintaining a simple metric. The procedure to do so is explained in the Appendix.

Appendix: stretched grid and metric, also time dependent

The number of grid points in x is M, with $0 \leq i \leq M-1$. The grid is stretched using a mapping $\tilde{i} = g(i)$ with $0 \leq \tilde{i} \leq 1$. The position of the grid points is thus obtained as

$$x_i = x(i) = L\,\tilde{i} - L_u = L\,g(i) - L_u \; . \tag{14}$$

The grid resolution is thus obtained as

$$\frac{dx}{di}(i) = L\,\frac{d\tilde{i}}{di}(i) = L\,g'(i) \; . \tag{15}$$

The stretching is here defined in 3 zones:

1. An upstream zone with stretching for $0 \leq i \leq i_1$. It covers the region from $-L_u \leq x \leq -\frac{c}{2}$: its span is thus $L_1 = L_u - \frac{c}{2}$. The mapping is

$$\tilde{i}(i) = g(i) = \frac{\left(L_u - \frac{c}{2}\right)}{L}\,\frac{\tanh\left(\beta_1\,\frac{i}{i_1}\right)}{\tanh(\beta_1)} \; . \tag{16}$$

   We obtain $\tilde{i}_0 = \tilde{i}(0) = 0$ and $\tilde{i}_1 = \tilde{i}(i_1) = \frac{\left(L_u - \frac{c}{2}\right)}{L}$, as it should. The grid resolution is obtained as

$$\begin{aligned}
\frac{dx}{di}(0) &= \left(L_u - \frac{c}{2}\right)\frac{1}{i_1}\,\frac{\beta_1}{\tanh(\beta_1)} \; , \\
\frac{dx}{di}(i_1) &= \left(L_u - \frac{c}{2}\right)\frac{1}{i_1}\,\frac{\beta_1}{\sinh(\beta_1)\,\cosh(\beta_1)} = \left(L_u - \frac{c}{2}\right)\frac{1}{i_1}\,\frac{2\,\beta_1}{\sinh(2\,\beta_1)} \; , \\
\frac{dx}{di}(i_1)/\frac{dx}{di}(0) &= \frac{1}{\cosh^2(\beta_1)} \; .
\end{aligned} \tag{17}$$

2. A middle zone without stretching for $i_1 \leq i \leq i_2$. It covers the region $-\frac{c}{2} \leq x \leq \frac{3\,c}{2}$: its span is thus $L_2 = 2\,c$. The mapping is

$$\tilde{i}(i) = g(i) = \tilde{i}_1 + \left(\tilde{i}_2 - \tilde{i}_1\right)\frac{(i - i_1)}{(i_2 - i_1)} = \frac{\left(L_u - \frac{c}{2}\right)}{L} + \frac{2\,c}{L}\,\frac{(i - i_1)}{(i_2 - i_1)} \; . \tag{18}$$

   We obtain $\tilde{i}_1 = \tilde{i}(i_1) = \frac{\left(L_u - \frac{c}{2}\right)}{L}$ and $\tilde{i}_2 = \tilde{i}(i_2) = \frac{\left(L_u + \frac{3\,c}{2}\right)}{L}$, as it should. The grid resolution is uniform and is obtained as

$$\frac{dx}{di}(i) = 2\,c\,\frac{1}{(i_2 - i_1)} \; . \tag{19}$$

3. A downstream zone with stretching. It covers the region from $\frac{3\,c}{2} \leq x \leq L - L_u = L_d + c$: its span is thus $L_3 = L - \left(L_u + \frac{3\,c}{2}\right) = L_d - \frac{c}{2}$. The mapping is

$$\tilde{i}(i) = g(i) = \frac{\left(L_u + \frac{3\,c}{2}\right)}{L} + \left(1 - \frac{\left(L_u + \frac{3\,c}{2}\right)}{L}\right)\left(1 - \frac{\tanh\left(\beta_3\left(1 - \frac{(i - i_2)}{((M-1) - i_2)}\right)\right)}{\tanh(\beta_3)}\right) \; . \tag{20}$$

We obtain $\tilde{i}_2 = \tilde{i}(i_2) = \frac{\left(L_u + \frac{3\,c}{2}\right)}{L}$ and $\tilde{i}_{M-1} = \tilde{i}(M-1) = 1$, as it should. The grid resolution is obtained as

$$\begin{aligned}
\frac{dx}{di}(i_2) &= \left(L - \left(L_u + \frac{3\,c}{2}\right)\right) \frac{1}{((M-1)-i_2)} \frac{2\,\beta_3}{\sinh(2\,\beta_3)} \,, \\
\frac{dx}{di}(M-1) &= \left(L - \left(L_u + \frac{3\,c}{2}\right)\right) \frac{1}{((M-1)-i_2)} \frac{\beta_3}{\tanh(\beta_3)} \,, \\
\frac{dx}{di}(i_2)/\frac{dx}{di}(M-1) &= \frac{1}{\cosh^2(\beta_3)} \,.
\end{aligned} \tag{21}$$

Important: To ensure a nice continuity of the resolution between the 3 zones, the following relations should be satisfied (at least approximately)

$$\begin{aligned}
\left(L_u - \frac{c}{2}\right) \frac{1}{i_1} \frac{2\,\beta_1}{\sinh(2\,\beta_1)} &= 2\,c\,\frac{1}{(i_2 - i_1)} \\
&= \left(L - \left(L_u + \frac{3\,c}{2}\right)\right) \frac{1}{((M-1)-i_2)} \frac{2\,\beta_3}{\sinh(2\,\beta_3)}
\end{aligned} \tag{22}$$

Thus, once i_1 and i_2 are chosen for some chosen $\frac{L}{c}$, $\frac{L_u}{c}$ and M, one can obtain proper values for β_1 and β_3.

The number of grid points in y for the upper part (= upper computational domain) is N, with $0 \leq j \leq N-1$. The grid is stretched using a mapping $\tilde{j} = h(j)$ with $0 \leq \tilde{j} \leq 1$. the mapping is:

$$\tilde{j}(j) = h(j) = 1 - \frac{\tanh\left(\beta_+ \left(1 - \frac{j}{(N-1)}\right)\right)}{\tanh(\beta)} \,. \tag{23}$$

We indeed obtain $\tilde{j}_0 = \tilde{j}(0) = 0$ and $\tilde{j}_{(N-1)} = \tilde{j}(N-1) = 1$.

We then further link $\tilde{j}$ to the geometry, with the channel upper wall located at $y = \frac{H}{2}$ and the middle geometry $f_{m^+}(x)$ corresponding to a horizontal line for $-L_u \leq x \leq 0$, connected to the curve representing the upper part of the airfoil for $0 \leq x \leq c$, and then connected to another horizontal line for $c \leq x \leq c + L_d = L - L_u$. We then have

$$\tilde{j} = \frac{(y - f_{m^+}(x))}{\left(\frac{H}{2} - f_{m^+}(x)\right)} \,. \tag{24}$$

When the airfoil has no angle of attack, the two horizontal lines are aligned. If the airfoil is also at the center of the channel, they are located at $y_c = 0$; if the airfoil is located instead at $y_c = \delta \frac{H}{2}$, then they are also located at y_c. When the airfoil is put at some angle of attack, the two horizontal lines are no longer aligned; yet, their mean position (average between the left and right lines) is still the center the airfoil, located at y_c.

We also obtain

$$\frac{\partial \tilde{j}}{\partial y} = \frac{1}{\left(\frac{H}{2} - f_{m^+}(x)\right)} \,,$$

$$
\begin{aligned}
\frac{\partial j}{\partial y} &= \frac{dj}{d\tilde{j}}\frac{\partial \tilde{j}}{\partial y} = \frac{1}{\frac{d\tilde{j}}{dj}}\frac{\partial \tilde{j}}{\partial y} = \frac{1}{h'(j)\left(\frac{H}{2} - f_{m^+}(x)\right)}\ , \\
\frac{\partial \tilde{j}}{\partial x} &= -\frac{f'_{m^+}(x)}{\left(\frac{H}{2} - f_{m^+}(x)\right)} + \frac{(y - f_{m^+}(x))}{\left(\frac{H}{2} - f_{m^+}(x)\right)}\frac{f'_{m^+}(x)}{\left(\frac{H}{2} - f_{m^+}(x)\right)} = -\frac{(1-h(j))\, f'_{m^+}(x)}{\left(\frac{H}{2} - f_{m^+}(x)\right)}\ , \\
\frac{\partial j}{\partial x} &= \frac{dj}{d\tilde{j}}\frac{\partial \tilde{j}}{\partial x} = \frac{1}{\frac{d\tilde{j}}{dj}}\frac{\partial \tilde{j}}{\partial x} = -\frac{(1-h(j))\, f'_{m^+}(x)}{h'(j)\left(\frac{H}{2} - f_{m^+}(x)\right)}\ .
\end{aligned}
\tag{25}
$$

When we simulate an airfoil with some angle of attack, or even an airfoil at zero angle attack but not located in the center of the channel, we need to use two computational domains that share the two horizontal lines mentioned above. Two domains are also required when we simulate unsteady problems: see later. In fact, the only case that could be simulated using solely the upper computational domain, with also enforcing up-down symmetry, is the case of an airfoil located in the center of the channel, at zero angle of attack, and without movement. We thus consider the general case in what follows, and we program the code using two computational domains.

The number of grid points in y for the lower part (= lower computational domain) is also taken as N, with $0 \le j \le N-1$. The grid is also stretched using a mapping $\tilde{j} = h(j)$ with $-1 \le \tilde{j} \le 0$. The mapping is:

$$
\tilde{j}(j) = h(j) = \frac{\tanh\left(\beta_- \frac{j}{(N-1)}\right)}{\tanh(\beta)} - 1\ . \tag{26}
$$

We indeed obtain $\tilde{j}_0 = \tilde{j}(0) = -1$ and $\tilde{j}_{(N-1)} = \tilde{j}(N-1) = 0$. The link of $\tilde{j}$ to the geometry is then

$$
\tilde{j} = \frac{(y - f_{m^-}(x))}{\left(\frac{H}{2} + f_{m^-}(x)\right)} \tag{27}
$$

with the channel lower wall located at $y = -\frac{H}{2}$ and the middle geometry $f_{m^-}(x)$ corresponding to a horizontal line (= same line as for the upper domain) for $-L_u \le x \le 0$, connected to the curve representing the lower part of the airfoil for $0 \le x \le c$, and then connected to another horizontal line (same line as for the upper domain) for $c \le x \le c + L_d = L - L_u$.

We also obtain

$$
\begin{aligned}
\frac{\partial \tilde{j}}{\partial y} &= \frac{1}{\left(\frac{H}{2} + f_{m^-}(x)\right)}\ , \\
\frac{\partial j}{\partial y} &= \frac{dj}{d\tilde{j}}\frac{\partial \tilde{j}}{\partial y} = \frac{1}{\frac{d\tilde{j}}{dj}}\frac{\partial \tilde{j}}{\partial y} = \frac{1}{h'(j)\left(\frac{H}{2} + f_{m^-}(x)\right)}\ , \\
\frac{\partial \tilde{j}}{\partial x} &= -\frac{f'_{m^-}(x)}{\left(\frac{H}{2} + f_{m^-}(x)\right)} - \frac{(y - f_{m^-}(x))}{\left(\frac{H}{2} + f_{m^-}(x)\right)}\frac{f'_{m^-}(x)}{\left(\frac{H}{2} + f_{m^-}(x)\right)} = -\frac{(1+h(j))\, f'_{m^-}(x)}{\left(\frac{H}{2} + f_{m^-}(x)\right)}\ , \\
\frac{\partial j}{\partial x} &= \frac{dj}{d\tilde{j}}\frac{\partial \tilde{j}}{\partial x} = \frac{1}{\frac{d\tilde{j}}{dj}}\frac{\partial \tilde{j}}{\partial x} = -\frac{(1+h(j))\, f'_{m^+}(x)}{h'(j)\left(\frac{H}{2} + f_{m^-}(x)\right)}\ .
\end{aligned}
\tag{28}
$$

Important: To ensure a nice continuity of the resolution at the connection of the upper and lower computational domains, the following relations should be satisfied (at least approximately):

$$\left(\frac{H}{2} - y_c\right) \frac{1}{(N-1)} \frac{2\,\beta_+}{\sinh(2\,\beta_+)} = \left(\frac{H}{2} + y_c\right) \frac{1}{(N-1)} \frac{2\,\beta_-}{\sinh(2\,\beta_-)}\,. \tag{29}$$

Cases of moving airfoils: to simulate moving airfoils, we must replace $f_{m^+}(x,)$ and $f_{m^-}(x,)$ by $f_{m^+}(x,t)$ and $f_{m^-}(x,t)$. To maintain a simple mapping, we decide to only deform the grid vertically. Thus, the shared horizontal lines maintain their length (that upstream of the airfoil ends at $x = 0$ and that downstream of the airfoil starts at $x = c$). We also restrict ourselves to small deformations, so that we don't have to vary β_+ and β_- in time.

When we study an airfoil that is oscillating vertically, the two curves move in a synchronous way: $f_{m^+}(x,t) = f_{m^+}(x) + d(t)$ and $f_{m^-}(x,t) = f_{m^-}(x) + d(t)$ with $d(t) = (\mu\, c)\sin(\omega\, t)$ with $\mu \ll 1$ the dimensionless amplitude and ω the circular frequency (= pulsation).

When we study an airfoil that is oscillating in pitch, the two curves move in an asynchronous way:

1. the shared horizontal line upstream of the airfoil moves according to $f_{m^+}(t) = f_{m^+}(t) = y_c + (\mu\, c)\sin(\omega\, t)$ with $\mu \ll 1$;
2. the shared horizontal line downstream of the airfoil moves according to $f_{m^+}(t) = f_{m^-}(t) = y_c - (\mu\, c)\sin(\omega\, t)$;
3. the part that describes the upper airfoil surface moves according to $f_{m^+}(x,t) = f_{m^+}(x) + \left(1 - 2\,\frac{x}{c}\right)(\mu\, c)\sin(\omega\, t))$ and the part that describes the lower airfoil surface moves according to $f_{m^-}(x,t) = f_{m^-}(x) + \left(1 - 2\,\frac{x}{c}\right)(\mu\, c)\sin(\omega\, t))$.

Finally, we can also study an airfoil that expands and contracts in time: this corresponds to modifying the parts of f_{m^+} and f_{m^-} that describe the upper and lower airfoil surfaces by $f_{m^+}(x,t) = f_{m^+}(x)\,(1 + \mu\,\sin(\omega\, t))$ and $f_{m^-}(x,t) = f_{m^-}(x)\,(1 - \mu\,\sin(\omega\, t))$ with $\mu < 1$; while not moving the shared horizontal line upstream and downstream.

Since the grid stretching in x depends only on i, and since the deformation of f_{m^+} and f_{m^+} only depends on i and t, the Jacobian matrix J^{-1} is much simpler than for a general case. It is obtained as:

$$\begin{aligned}
J^{-1} &= \begin{pmatrix} 1 & 0 & 0 \\ \frac{\partial i}{\partial t} & \frac{\partial i}{\partial x} & \frac{\partial i}{\partial y} \\ \frac{\partial j}{\partial t} & \frac{\partial j}{\partial x} & \frac{\partial j}{\partial y} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{di}{dx} & 0 \\ \frac{\partial j}{\partial t} & \frac{\partial j}{\partial x} & \frac{\partial j}{\partial y} \end{pmatrix} \\
&= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{L\,g'(i)} & 0 \\ -\frac{(1 \mp h(j))\,\frac{\partial f_{m^\pm}}{\partial t}(i,t)}{h'(j)\left(\frac{H}{2} \mp f_{m^\pm}(i,t)\right)} & -\frac{(1 \mp h(j))\,\frac{\partial f_{m^\pm}}{\partial x}(i,t)}{h'(j)\left(\frac{H}{2} \mp f_{m^\pm}(i,t)\right)} & \frac{1}{h'(j)\left(\frac{H}{2} \mp f_{m^\pm}(i,t)\right)} \end{pmatrix}
\end{aligned} \tag{30}$$

and its determinant is simply $\mathcal{D}^{-1} = \frac{di}{dx}\,\frac{\partial j}{\partial y} = \frac{1}{L\,g'(i)\,h'(j)\left(\frac{H}{2} \mp f_{m^\pm}(i,t)\right)}$.

The Jacobian matrix is thus

$$\begin{aligned} J &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{dx}{di} & 0 \\ \frac{\partial y}{\partial t} & \frac{\partial y}{\partial i} & \frac{\partial y}{\partial j} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{dx}{di} & 0 \\ -\frac{\frac{\partial j}{\partial t}}{\frac{\partial j}{\partial y}} & -\frac{dx}{di}\,\frac{\frac{\partial j}{\partial x}}{\frac{\partial j}{\partial y}} & \frac{1}{\frac{\partial j}{\partial y}} \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & L\,g'(i) & 0 \\ (1 \mp h(j))\frac{\partial f_{m^\pm}}{\partial t}(i,t) & L\,g'(i)\left((1 \mp h(j))\,\frac{\partial f_{m^\pm}}{\partial x}(i,t)\right) & h'(j)\left(\frac{H}{2} \mp f_{m^\pm}(i,t)\right) \end{pmatrix} \end{aligned} \tag{31}$$

and its determinant is $\mathcal{D} = L\,g'(i)\,h'(j)\left(\frac{H}{2} \mp f_{m^\pm}(i,t)\right)$.